ENGR 111
Introduction to Engineering
Class #14

Measurement and error

Ten readings of the same thing give you ten numbers.

Zoom protocol

Name Set your Zoom name to your real name, or whatever you'd like me to call you.

Photo Set a professional profile photo.

Camera Encouraged, not required.

Mic Off unless you're speaking.

Your questions, anytime Type in chat or raise your hand, and I'll check periodically. Or just unmute and jump in.

My questions to the class Answer in chat, or unmute and answer.

If you speak Please state your name first.

If I drop out (e.g., my Wi-Fi fails) Stay in the meeting. I'll be back in a few minutes on my phone hotspot.

How many ping-pong balls can fit in a room

An estimate is arithmetic you can show someone. Order of magnitude, not decimal places.

Fast: pack them in cubes

Room: $10 \times 15 \times 10$ ft $= 1{,}500$ ft³

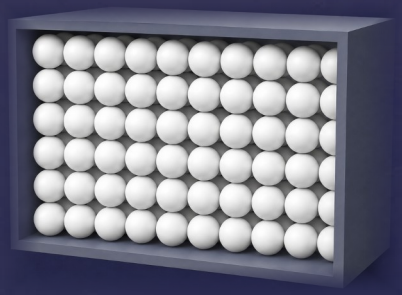
Ball: 1.5 in across, so 0.125 ft

Each ball gets its own cube: $0.125^3 = 0.001953$ ft³

A ball fills $\pi/6 = 52\%$ of its cube.

The corners are air, and you just counted them as ball.

$1{,}500 \div 0.001953 = 768{,}000$



Slower: use a packing factor

One ball: $(4/3)\pi r^3 = 0.001023$ ft³ of plastic

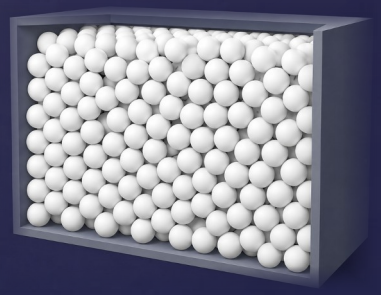
Nestle them into each other's gaps instead.

Stacked spheres fill about 70% of a box.

Looked up, not obvious. The proven best is 74%.

$1{,}500 \times 0.7 = 1{,}050$ ft³ ball, 450 ft³ air

$1{,}050 \div 0.001023 = 1{,}026{,}740$



Two methods, 34% apart, and both say about a million.
Same order of magnitude. That is all an estimate promises.

Measurement error and variation

A measurement error is the measured value minus a reference value. The reference may be a known standard or the unknown true value, so the exact error is usually unknown. Repeated readings can also differ from one another.

$$\text{measurement error} = \text{measured value} - \text{reference value}$$

Four common sources are:

Instrument. The device. A scale may round to the nearest gram, have a zero offset, or drift.

Operator. The person taking the measurement. Reaction time, viewing angle, and technique can affect the result.

Environmental. The surrounding conditions. Temperature, vibration, humidity, or electrical noise can affect the result.

Object or event. The thing being measured may genuinely change from one trial to the next.

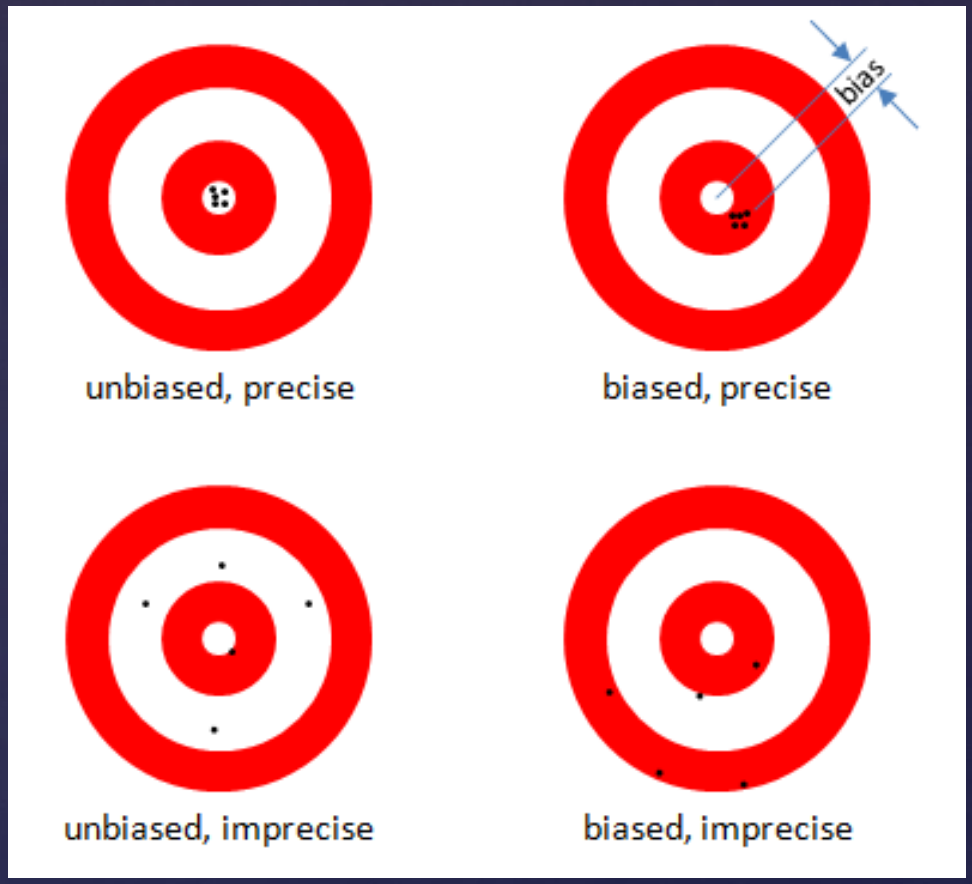
Some effects push every reading the same way. Others make the readings scatter.

Bias, precision, and accuracy

Bias: Offset between average and the reference value.

Precision: How closely repeated readings agree (the spread).

Accuracy: Overall closeness to the reference value being measured.



Bullseye = trusted reference value

Good accuracy requires small bias and high precision.

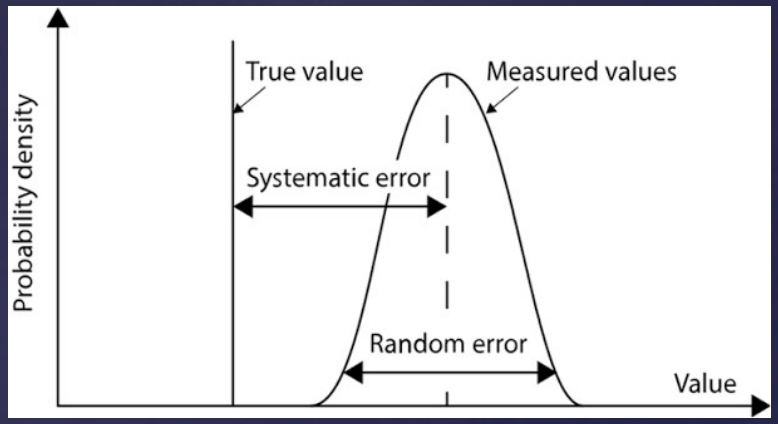
What causes bias and scatter?

Systematic error produces bias.

Random error produces scatter and limits precision.

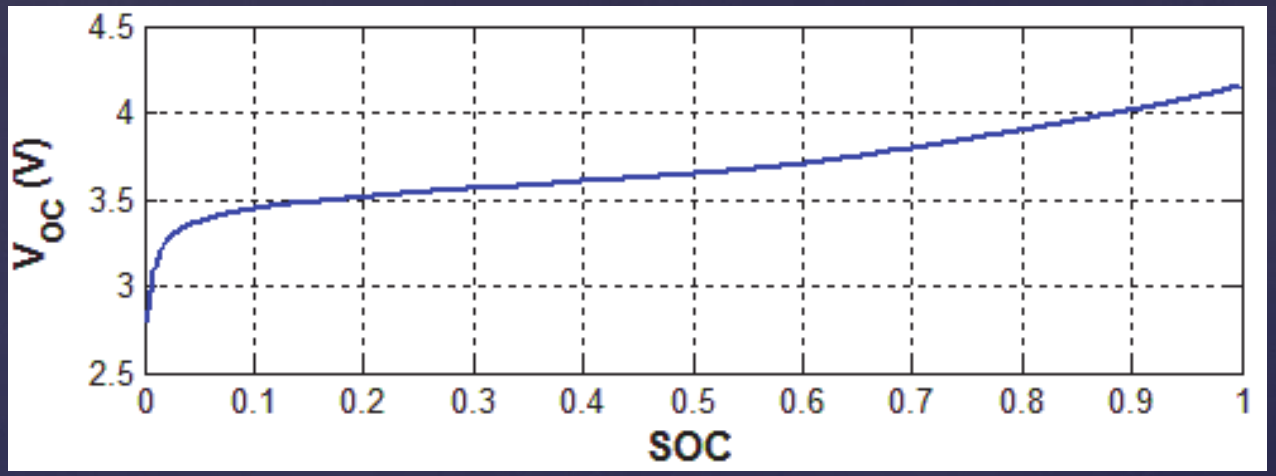
Calibration addresses bias.

Repetition and averaging reduce random error in the mean.



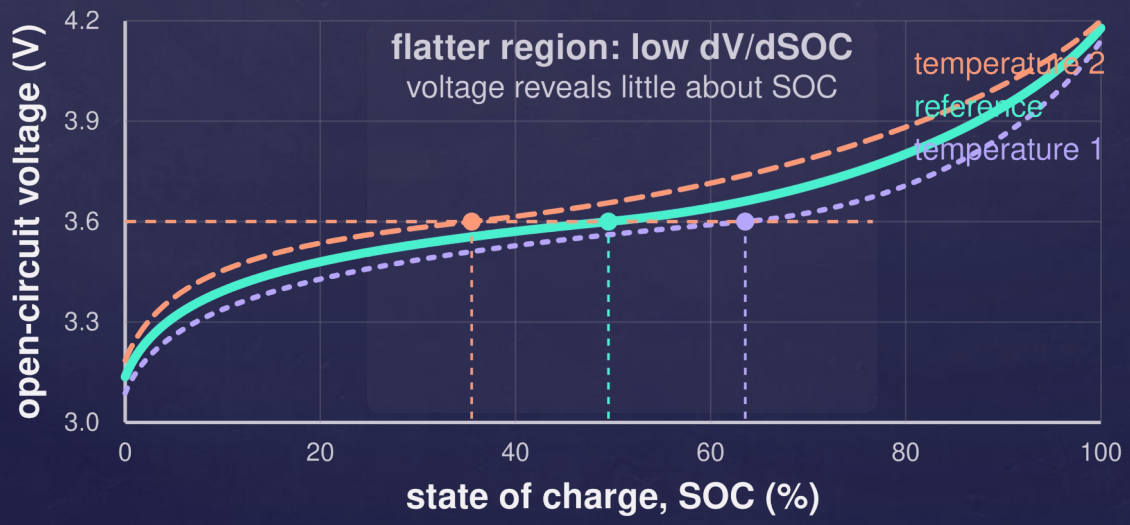
Sensitivity

Sensitivity is the change in an instrument's output divided by the change in a particular input. Desired sensitivity provides information; unwanted cross-sensitivity can introduce error.



We use the sensitivity of voltage to state of charge (SOC) to estimate the battery SOC (fuel gauge)





Desired
SOC changes voltage

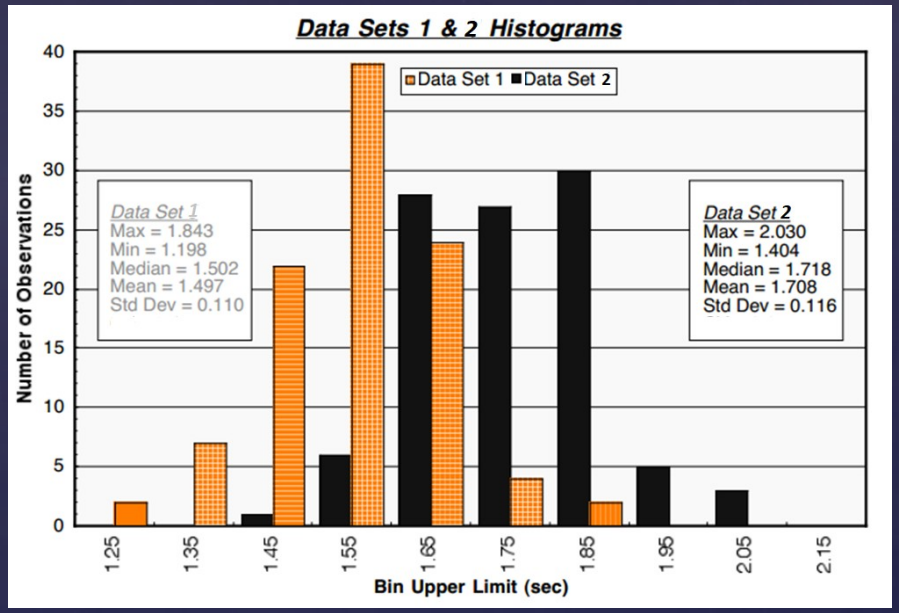
Unwanted
temperature, current, and aging also change voltage

SOC error
unless controlled or compensated

Two data sets, one true value

The same 1.700 second interval, measured 100 times on each of two instruments.

What a histogram is. Readings get sorted into bins, and each bar counts how many landed in that bin.
Two instruments, one quantity. Orange is one of them, black is the other.
The boxes do the arithmetic. Mean and standard deviation for each set are printed on the chart.



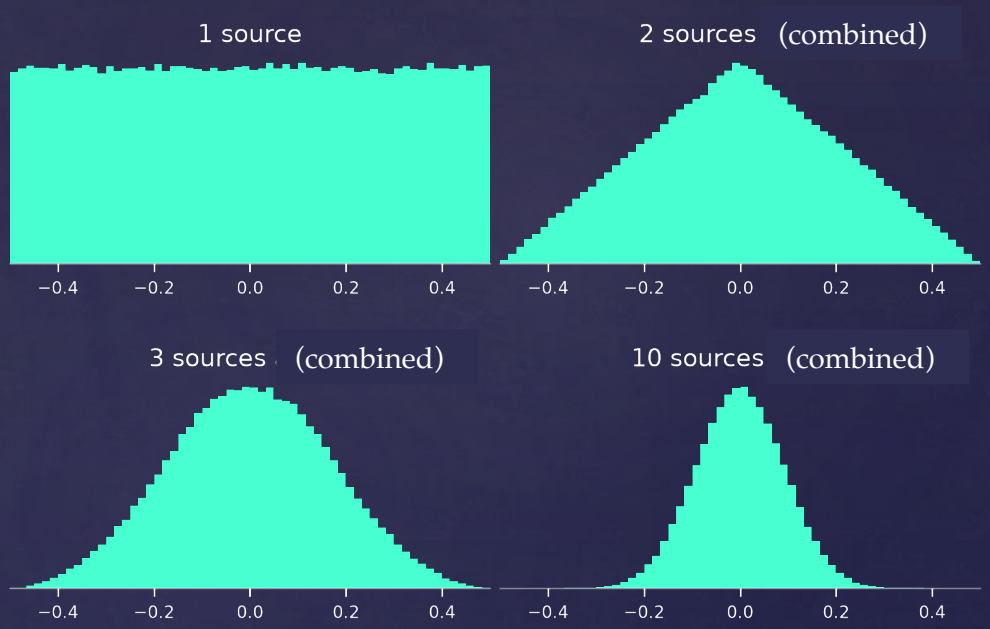
Which data set (orange vs black) has a larger bias? Larger accuracy? Larger precision?

Why measurement noise so often forms a bell curve

A single noise source can have almost any distribution. But combine many independent noise sources, and their total tends toward a bell curve.

source A	source B	(A + B)/2
+0.42	-0.31	+0.055
-0.48	+0.45	−0.015
+0.07	+0.12	+0.095
-0.22	+0.30	+0.040
+0.44	+0.47	+0.455
-0.11	-0.06	−0.085
⋮	⋮	⋮

Rare: both large and in the same direction



Horizontal scale normalized to compare shape

The surprise: The bell curve does not have to be present in any individual source. It emerges from adding many independent random effects. This math theorem behind this is called the Central Limit Theorem, and explains why bell curves appear so often in measurement noise…and throughout nature.

Mean and standard deviation

Ten stopwatch readings of one event, in seconds.

12.41	12.28	12.55	12.33	12.47
12.19	12.62	12.38	12.44	12.31

Mean = average

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = (12.41 + 12.28 + \ldots + 12.31) \div 10 = 12.398 \text{ s}$$

Standard deviation = scatter. How much individual readings typically vary from the mean.

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2} = \sqrt{[\,((12.41 - 12.398)^2 + \ldots + (12.31 - 12.398)^2) \div 9\,]} = 0.129 \text{ s}$$

The measurements are centered near 12.40 s, with a typical scatter of about 0.13 s

Why n − 1, not n. Because the mean was estimated from these same measurements. Using n-1 prevents us from underestimating the scatter.

Any tool does this. =AVERAGE() and =STDEV.S() in a spreadsheet, or paste the ten readings into an AI and ask for the mean, the standard deviation, and a histogram.

Don’t report only the mean, report how well you know it

Repeated measurements provide one way to estimate uncertainty caused by random scatter.
From previous slide:

$n = 10, \quad \bar{x} = 12.398\ \text{s}, \quad s = 0.129\ \text{s}$

95% confidence interval = $\bar{x} \pm t\frac{s}{\sqrt{n}} = 12.398 \pm 2.262\frac{0.129}{\sqrt{10}} = 12.398 \pm 0.092\ \text{s}$

For ten readings (n = 10), the 95% Student’s t multiplier is 2.262.

$12.31\ \text{s} < \text{true mean} < 12.49\ \text{s}$

We are 95% confident that the true mean under these measurement conditions lies between 12.31 s and 12.49 s.

What 95% confidence means. Imagine repeatedly taking 10 measurements and calculating a new interval each time. About 19 out of 20 (95%) of those intervals would contain the fixed true mean.

→ More independent measurements reduce random uncertainty, but they do not remove systematic bias.

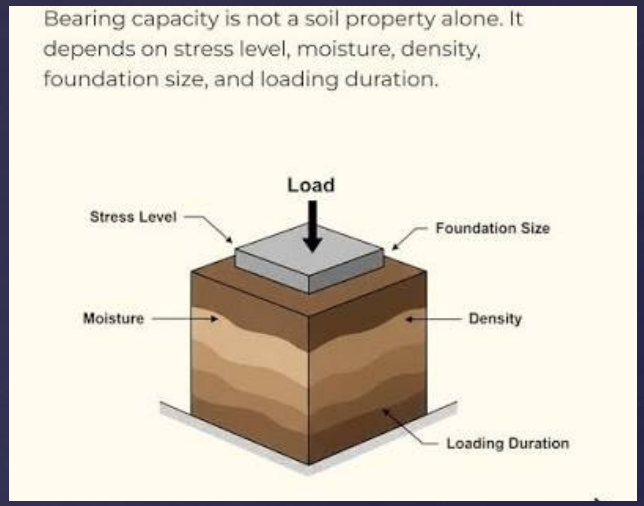
Assumes independent readings with approximately normally distributed random measurement error.

From measurement uncertainty to a safe design

A foundation spreads a building's weight over the soil. If the pressure is too high, the soil settles unevenly, potentially cracking walls and jamming doors.

Bearing capacity is the pressure the soil can safely support.

Soil varies with location, moisture, composition, and compaction. Boreholes test only a few locations, so the capacity between them remains uncertain.



Suppose soil engineer reports allowable bearing capacity of $q = 2{,}500\ \mathrm{lb/ft^2} \pm 50\%$

$$1{,}250 \le q \le 3{,}750\ \mathrm{lbf/ft^2}$$

Design formula:

$$\text{Foundation area A} = \frac{\text{building weight}}{\text{bearing capacity}}$$

Building weight $W = 2.14 \times 10^6$ lbf

QUESTION: You are the engineer. What foundation area A would you recommend, and how does the uncertainty in q affect your decision?

Note: Here the ±50% is a hypothetical engineering uncertainty range (representing limited sampling, changing conditions, and engineering judgment) not a 95% confidence interval or previous Student's t calculation.